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Course: ITAI-1378-Comp Vision-Artificial Intel

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Image Processing Fundamentals

Section 1: Technical Understanding

The most surprising discovery was learning that computers see images as grids of numbers called matrices. When I separated the Red, Green, and Blue channels, I saw how a color picture breaks down into three simple layers. Every pixel is just a number from 0 to 255.

Simple math creates the visual effects we see on screens. Changing brightness means adding numbers to every pixel. Changing contrast means multiplying pixel values. When we used a 3x3 kernel, it acted like a tiny window sliding across the image to calculate new values. This neighborhood math lets us blur images or find sharp edges.

The most difficult technique to understand was CLAHE. Unlike regular adjustments that change the whole image, CLAHE fixes contrast inside small squares called tiles. It was hard to understand how the math clips high points to stop noise and blends the tiles together smoothly.

Section 2: Connections and Applications

This lab connects directly to the Nano Banana computer vision tool from class. Nano Banana finds and labels objects using convolution filters just like the 3x3 kernels we made. The main difference is that we chose our filter numbers by hand, while Nano Banana automatically learns the best numbers from training data.

These traditional techniques are highly useful for e-commerce sites. If a store has one million product photos, using AI for everything is too slow and expensive. Instead, fast traditional scripts can instantly fix lighting, crop borders, and convert files to WebP to save money and speed up page load times.

In the future, I would combine traditional image processing with AI tools. I can use traditional filters like CLAHE as a preprocessing step to clean dark or blurry images first. Making the data clean and sharp beforehand makes it much easier for a deep learning AI model to accurately classify objects.

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Section 3: Personal Reflection

I am most interested in exploring custom filter design. I want to learn how changing numbers in a kernel matrix can isolate specific details, like horizontal lines or textures, and how smartphone cameras use these filters automatically.

This lab changed my view of digital photography and editing software. I used to think phone filters were just creative magic. Now I realize that tools like Photoshop are just fast matrix math. Moving an editing slider simply changes numbers in a pixel grid.

I am also very curious about image security and privacy. This lab taught me how to read raw pixel data, which made me wonder how engineers protect sensitive photos (like medical images or user data) from being hacked or changed. I want to learn how we can use matrix math to encrypt images or hide secret data inside pixels.